

Improving SAM-Road Model for Occlusion Handling in Road Networks Extraction from Satellite Images with Gamma Correction and Modified A* Algorithm

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Abstract

Background: Occlusion in satellite imagery often leads to disconnected road networks, reducing the quality and reliability of geospatial data, which in turn hampers infrastructure and transportation planning. Although models like SAM-Road show promising results, they still struggle in handling occluded areas, especially at intersections and curved roads. Some existing methods, such as the extended-line approach, have been proposed to address occlusion; however, they are typically limited to handling linear road segments and are less effective in complex structures. To overcome these limitations, this study enhances the SAM-Road framework by incorporating gamma correction and a modified A algorithm in the post-processing stage. This combined approach improves the visibility of partially hidden roads and successfully reconnects fragmented segments, even in non-linear and occlusion-heavy areas.

Objective: This study aims to improve the accuracy of road network extraction by enhancing the SAM-Road model with gamma correction and a modified A* algorithm, specifically to address the problem of occlusion in satellite imagery.

Methods: This research adopts a quantitative experimental approach. Gamma correction is applied to enhance the visual contrast of roads in occluded satellite images, while the A pathfinding algorithm is modified to reconnect disjointed road segments. The integrated method is then evaluated using accuracy metrics, specifically the TOPO and APLS (Average Path Length Similarity) scores.

Results: The experimental findings indicate that each method—SAM-Road baseline, gamma correction, the modified A* algorithm, and their combination—delivers distinct performance improvements. Gamma correction alone achieves the best results at gamma 1.5 (TOPO 80.61%) and 1.25 (APLS 70.91%) on SpaceNet, and gamma 2.0 (TOPO 78.56%) and 1.25 (APLS 68.73%) on City-scale. The modified A* algorithm performs best at 16/8 (TOPO 80.22%) and 32/16 (APLS 70.71%) on SpaceNet, and 16/8 (TOPO 77.29%) and 64/32 (APLS 70.94%) on City-scale. When combined, the method yields results within the range of TOPO 75.41–80.59% and APLS 67.59–71.17% on SpaceNet, and TOPO 76.52–78.56% and APLS 66.39–71.19% on City-scale.

Conclusion: This study concludes that the integration of gamma correction and a modified A* algorithm effectively addresses occlusion-related challenges in satellite imagery. While each technique contributes unique improvements, their combination significantly enhances the accuracy and continuity of extracted road networks—not only in straight road segments such as extended-line method, but also in more complex or occluded areas. The results confirm that this hybrid approach yields road extraction outputs that more closely align with ground truth in terms of topology and structure. Future research could explore integrating gamma correction and the modified A* algorithm directly into the training process, aiming to enhance model performance while maintaining high accuracy.

Keywords: SAM-Road, modified A*, gamma correction, occlusion, road network extraction, satellite imagery, topology, geometry

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I. INTRODUCTION

In recent years, developing countries have faced significant challenges in managing urban mobility due to the rising ownership of private vehicles, rapid economic growth, urbanization, and population increases [1]. These challenges have led to severe traffic congestion, higher fuel consumption, and strain on existing transportation infrastructure and management [2]. Among the most impacted sectors is urban road networks, which play a central role in supporting the movement of people and facilitating economic activities [3], [4].

Given the constraints on transportation infrastructure, the need for efficient road management and planning systems

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has become increasingly urgent [5]. Various approaches have been explored to enhance road network efficiency, including the use of accurate spatial modeling. Such modeling allows for a more comprehensive evaluation of road network performance, which can, in turn, lead to more efficient solutions regarding travel costs, sustainability, and overall impact [6], [7].

One of the critical aspects in addressing these challenges is the use of geospatial data. Geospatial data provides essential information representing the location and physical characteristics of the Earth's surface, including infrastructure like road networks [8], [9]. This data is crucial for urban planning, transportation management, disaster management, and sustainable development. In the context of road networks, geospatial data allows for mapping, analysis, and decision-making that can significantly enhance planning processes [10]. However, the availability of accurate and complete geospatial road data remains a major challenge, particularly in developing countries [11], [12]. These challenges include the high costs associated with collecting detailed geospatial data—especially for densely populated urban areas—and the technological limitations in many regions. For instance, in Indonesia, the cost of producing topographic maps at a 1:5000 scale can reach up to \$7,988 USD per sheet, with the national budget for such projects estimated at \$2.99 billion USD [13].

Moreover, manual processes, such as the digitization of road maps, present their own set of challenges, including time consumption, a high dependence on human accuracy, and susceptibility to subjective errors [14]. These challenges are compounded by the uneven distribution of aerial-based technologies and geospatial data processing software across regions [15]. As a result, road network planning and management processes tend to be slow and inefficient, directly affecting the effectiveness of infrastructure planning for sustainable urban growth [16]. On the other hand, the increasing availability of high-resolution satellite imagery has spurred the need for automatic methods that are both fast and accurate for extracting road networks [17], offering a solution to the limitations of manual methods.

A notable advancement in this area is the introduction of the SAM-Road model, an adaptation of the Segment Anything Model (SAM) designed to extract large-scale road networks from satellite imagery [18]. SAM-Road represents the state-of-the-art in road network extraction based on satellite images, integrating two main phases: road segmentation and the construction of a road network graph from the segmentation results [18]. SAM-Road is a machine learning model that operates in two main cycles: training and inference. During training, the model learns patterns from labeled data to optimize its internal weights [19]. Pre-processing techniques such as image rotation (by 0°, 90°, 180°, or 270°) are used to augment the dataset without post-processing. In contrast, inference involves using the trained model to predict segmentations on new data, without pre-processing or additional post-processing. The output of this model is not just a segmentation image, but a road network representation in the form of a graph, where key points such as intersections are represented as nodes, and the road segments linking them are represented as edges.

The model's performance is evaluated using two primary metrics: the TOPO F1-score and Average Path Length Similarity (APLS) [20]. Previous studies have demonstrated that SAM-Road performs well on datasets like City-scale and SpaceNet, with TOPO F1-scores of 77.17% and 80.28%, and APLS of 67.61% and 70.68% for these datasets, respectively. Despite these promising results, SAM-Road faces real-world challenges, particularly in cases of occlusion, where areas of the road network are obscured by objects such as buildings, trees, or shadows. This results in disconnected road networks in the model's output [21], leading to incomplete or inaccurate topological representations.

Occlusion in satellite images can make road segmentation masks unclear or incomplete, which makes it harder to detect the full road network [22]. One way to help with this problem is by improving the visibility of the image before segmentation. Some studies have shown that gamma correction can enhance image details and contrast [23], which may help make blurry or partially hidden roads easier to detect. However, for cases where roads are entirely missed due to occlusion, the use of modified algorithms, such as the A* algorithm, has been suggested to reconnect disjointed road segments [24]. While SAM-Road++ has introduced the extended-line method to address such issues, it is primarily effective for straight road segments and less so for complex intersections or turns [21].

This study aims to improve the accuracy of the SAM-Road model by enhancing the quality of road network extraction in the presence of occlusion. This is achieved through the application of gamma correction and a modified A* algorithm [25], during the post-processing stage. This approach not only addresses straight road occlusion but also handles more complex areas, such as intersections and curves. To evaluate the effectiveness of this approach, two datasets are used namely SpaceNet and City-scale [26], [21]. These datasets also served as benchmarks for previous SAM-Road evaluations. This study will provide insights into the improvements in road network extraction and its applicability to urban planning, transportation analysis, and infrastructure management in developing countries.

II. METHODS

This study employs a quantitative research approach with an experimental design to evaluate the performance of the SAM-Road model, which is enhanced through the integration of gamma correction and a modified A algorithm, in improving the accuracy of road network extraction from satellite imagery. The overall experimental workflow proposed in this study is illustrated in Figure 1.

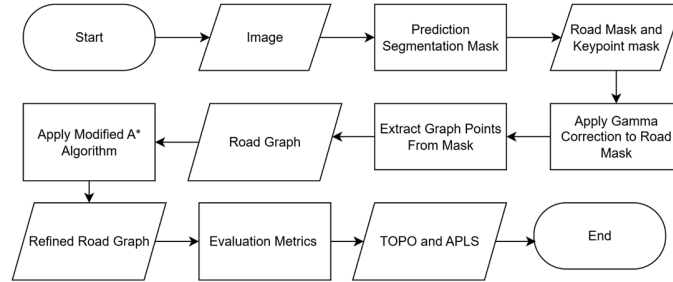


Fig. 1 The proposed system workflow.

A. Research Data

The data used in this study were obtained from two datasets, namely the SpaceNet dataset and the City-scale dataset [24]. Both datasets were employed to evaluate the performance of the SAM-Road model, particularly in constructing and assessing the quality of road network graphs extracted from satellite imagery. These datasets consist of collections of high-resolution satellite images depicting geographic areas with existing road networks. The images serve as the basis for the road network extraction process.

In this study, the evaluation of road network extraction models was carried out exclusively using the testing subsets of two widely recognized benchmark datasets, namely SpaceNet and City-scale. This choice was made because the focus of the study is solely on the inference stage, without any model training involved. The testing sets provide standardized and complete ground truth annotations required for reliable evaluation.

The SpaceNet dataset, which is sourced from DigitalGlobe's WorldView-3 satellite, consists of 2,549 high-resolution satellite images covering four diverse urban areas, namely Las Vegas, Paris, Shanghai, and Khartoum [27]. However, only the testing subset was utilized in this research, consisting of 382 images (approximately 15 percent of the total dataset). Each image represents a $0.4 \text{ km} \times 0.4 \text{ km}$ area with a 1-meter spatial resolution and dimensions of 400×400 pixels.

Similarly, the City-scale dataset, which covers a total area of 720 km^2 across 20 cities in the United States, was also used solely in its testing portion. Ground truth road networks for this dataset were derived from OpenStreetMap, while the corresponding satellite imagery was obtained using the Google Static Map API [28]. From the total of 180 images, this study employed only the 27 images designated for testing (also 15 percent of the dataset). Each image covers a $2 \text{ km} \times 2 \text{ km}$ area at 1-meter resolution with a size of 2048×2048 pixels.

By relying only on the testing data from each dataset and evaluating them separately, this study aims to provide a fair and accurate assessment of the model's ability to perform well on different urban areas and new satellite images. Examples of the data are shown in Figure 2.

B. Gamma Correction Method

In the road segmentation process using the SAM-Road model, visibility issues are frequently encountered in the resulting segmentation masks due to occlusions such as trees, buildings, and shadows [21]. These occlusions cause certain road areas to appear dimmer than others, making them difficult for the model to recognize.

Gamma correction is a technique in digital image processing used to adjust the relationship between pixel values in a digital image and the actual perceived brightness [29]. Gamma correction is applied to enhance the segmentation mask results produced by the prediction model, aiming to improve the clarity of the segmented road structures.

The use of a gamma value greater than 1 can result in a darker image, whereas a gamma value less than 1 produces a brighter image [30]. In this context, an inverse gamma approach is employed to ensure that the applied gamma value remains below 1. This is because a gamma value greater than 1 would darken the image, which contradicts the objective of enhancing image brightness [30].

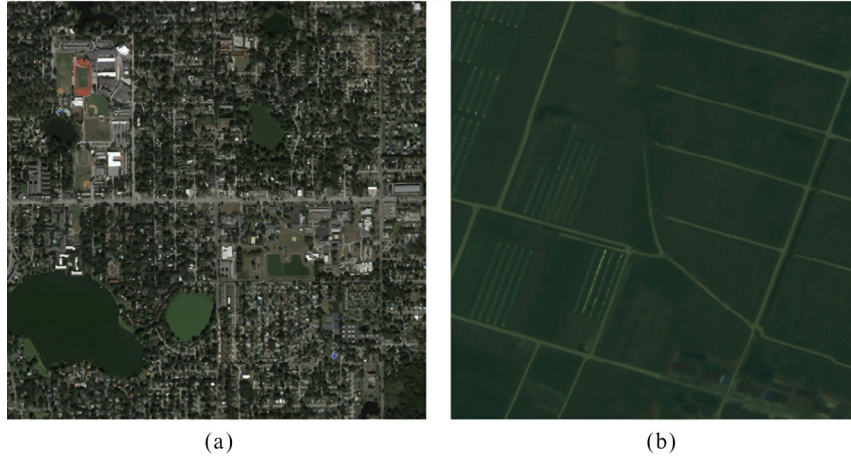


Fig. 2 An examples of dataset (a) City-scale and (b) SpaceNet.

The gamma value itself is a parameter that controls the brightness and contrast of the image after correction is applied. An excessively high gamma value may lead to over-enhancement, a condition where the image brightness increases excessively and introduces noise that degrades segmentation quality [31]. Therefore, selecting an appropriate gamma value is essential to ensure that the image becomes not only brighter but also retains important details crucial for the segmentation process. The gamma correction formula used in this study is presented in Equation (1).

$$f_0(x, y) = c \cdot \left(\frac{f_i(x, y)}{c} \right)^{1/\gamma} \quad (1)$$

Where,

- $f_0(x, y)$: Pixel intensity after gamma transformation
- $f_i(x, y)$: Pixel intensity before gamma transformation
- C : constant
- γ : Gamma value

To address this issue, a method is required that can enhance the clarity of predicted areas affected by occlusion, particularly within segmentation masks that exhibit low pixel intensity due to model uncertainty. Gamma correction was selected because this technique is capable of enhancing contrast in regions with low pixel intensity without significantly altering well-segmented areas. By clarifying the road segments in previously dim areas, the segmentation results become more distinct and suitable for the subsequent road network extraction process [29]. Figure 3 illustrates how gamma correction affects pixel intensity. The arrow in Figure 3(a) points to a road segment affected by occlusion, while Figure 3(b) shows the result after applying gamma correction.

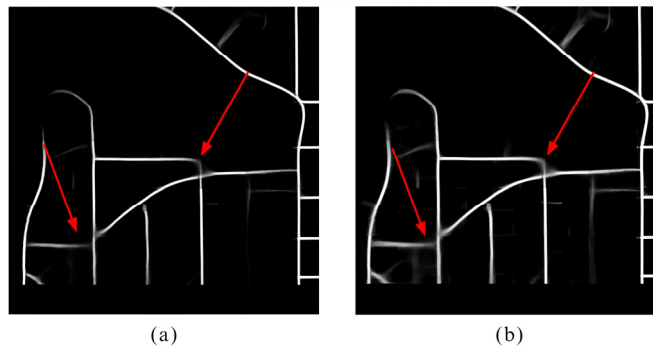


Fig. 3 Comparison of the segmentation mask (a) before and (b) after gamma correction.

Figure 4 illustrates the workflow of the gamma correction method applied to enhance the visibility of road segmentation masks generated by the SAM-Road model. The process begins with the predicted mask and a predefined

gamma value as input. First, the predicted mask is normalized by dividing each pixel intensity value by 255, thereby converting the range from [0, 255] to [0, 1]. This normalization step ensures that all pixel values are within a standardized range suitable for mathematical transformation. Next, gamma correction is applied using the inverse gamma operation, where each pixel value is raised to the power of $1/\gamma$ and then multiplied by 255 to return it to the original scale. This adjustment increases the brightness of dark regions while maintaining the clarity of already bright areas. As a result, the gamma-corrected mask exhibits improved contrast and visibility, especially in regions affected by occlusion such as shadows, trees, or buildings. The enhanced mask produced through this process serves as a more reliable input for subsequent road network extraction and graph construction stages.

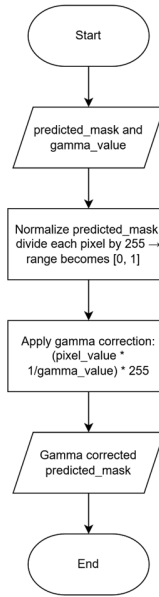


Fig. 4 Gamma Correction Method.

C. Modified A* Algorithm

A* is a shortest path search algorithm that operates by selecting paths based on heuristic cost estimates to determine the next node to visit, thereby making the search process more efficient. If the heuristic used is sufficiently accurate, the algorithm can find the optimal path more quickly [32]. The cost values considered in the A* algorithm include node cost, edge cost, and the heuristic function [33]. In this study, the A* algorithm is employed to improve road network connectivity by reconstructing connections between nodes that are disrupted due to occlusion in satellite imagery. A* is an extension of Dijkstra's algorithm, which is also designed to find the shortest path in a graph. The main difference between Dijkstra's algorithm and A* lies in the inclusion of a heuristic function in A* [34].

In the process of road network extraction using the SAM-Road model, a shortest pathfinding algorithm is employed as a post-processing step to address the issue of disconnected node connections. This issue is typically caused by occlusions in satellite imagery, which result in undetected road segments [21]. Although SAM-Road incorporates topology prediction based on a graph neural network [35], the presence of occlusions can still lead to incomplete road network extractions, leaving some nodes improperly connected. By applying a shortest pathfinding algorithm, missing connections between nodes can be reestablished based on the structure of the existing graph.

In this context, the A* algorithm is used with specific modifications adopted from the [33] paper, in which a reference point is introduced as a heuristic function to constrain the search space. The goal node concept is replaced by a search direction limitation, namely away from the reference point (opposite). In this study the modified A* algorithm incorporates two critical parameters: minimum graph distance and maximum straight distance. The minimum graph distance ensures that new connections are not formed between nodes that are too close in the existing graph structure, while the maximum straight distance limits the spatial extent within which potential connections can be searched. These parameters are complementary and directly proportional; in typical implementations, the minimum graph distance is set to be twice the maximum straight distance [33]. Therefore, the selection of these interval values follows previous studies to prevent the algorithm from generating connections between a node and a segment or another node that do not contribute to topological improvements in the road network structure.

Figure 5 illustrates how the modified A* algorithm can propose new candidate paths to improve road connectivity. A minimum graph distance (red box) is used to filter out connections that are too close, while a maximum straight distance (green box) defines the allowable search area. A reference point is marked in green, and the yellow line represents a candidate path connecting disjoint segments.

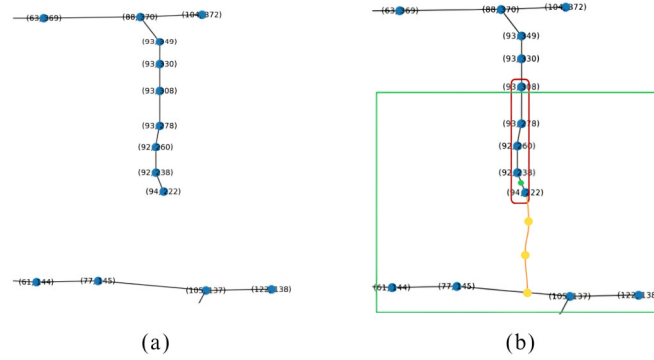


Fig. 5 An examples visualization of (a) original road graph and (b) new nodes and edges on the proposed path.

The Modified A* algorithm is designed to reconnect disconnected road segments within the extracted road graph by incorporating spatial and topological constraints. The algorithm begins by defining a search window centered at the starting pixel, where the search area is limited by the MAX_STRAIGHT_DISTANCE parameter to ensure that exploration remains within a relevant spatial range. The MIN_GRAPH_DISTANCE parameter is introduced to control the minimum allowable distance between the current pixel and the existing graph, preventing the formation of overly short or redundant connections. During the search process, the algorithm expands the pixel with the lowest total cost, combining movement cost, heuristic estimation, and segmentation-based penalties. The search continues iteratively until a valid connection path is found that satisfies the graph distance condition, after which the new path is reconstructed and integrated into the road graph to improve overall connectivity. The pseudocode of this algorithm can be seen in Algorithm 1.

Algorithm 1

Modified A* Algorithm

```

function modified Astar (start_pixel, opp, segmentation_mask, road_graph, MAX_STRAIGHT_DISTANCE,
MIN_GRAPH_DISTANCE)
  define search_window centered at start_pixel
    with radius based on MAX_STRAIGHT_DISTANCE
  initialize open_list with start_pixel
  initialize cost_map and path_trace
  while open_list is not empty do
    current_pixel ← pixel with lowest total_cost
    if current_pixel is on valid road edge
      and satisfies MIN_GRAPH_DISTANCE then
        reconstruct connection_path
        return connection_path
    end if
    for each neighbor_pixel in search_window do
      if neighbor_pixel is closer to start_pixel than to opp then
        compute movement_cost
        apply segmentation_penalty
        update path if lower total_cost found
      end if
    end for
  end while
  return "no valid path found"
end function

```

D. Definition of Baseline Metrics

The authors of SAM-Road utilized a Graphics Processing Unit (GPU) with 24 GB of VRAM for model training. However, this study was conducted in a constrained environment using the Google Colab platform, which only provides GPUs with 16 GB of VRAM. Due to this limitation, retraining the model was not feasible. Therefore, this study employed the pre-trained SAM-Road model, which had been previously trained on the SpaceNet and City-scale datasets by the original authors and made publicly available in a ready-to-use form. By utilizing the pre-trained model, the main focus of this research was directed toward the inference cycle i.e., the process in which the trained model is applied to new satellite imagery to generate road segmentation. This approach is considered valid as long as all inference procedures are conducted within the same environment [36].

In order to evaluate the performance of the proposed method, an appropriate baseline was established. The baseline was determined based on the use of this pre-trained model under consistent inference settings, ensuring a fair and reliable comparison across all evaluated scenarios. Specifically, the baseline values were obtained from inference results generated using a 16 GB VRAM GPU, and the same hardware environment was consistently maintained throughout all stages of this study. The resulting baseline values, as presented in Table 1, serve as the primary reference for evaluating the proposed approach.

TABLE 1
BASELINE PERFORMANCE METRICS OF THE PRE-TRAINED SAM-ROAD BASELINE [18] EVALUATED ON SPACENET AND CITY-SCALE DATASETS

Dataset	Model Configuration	TOPO F1-score	APLS
SpaceNet	SAM-Road Baseline [18]	80.28	70.68
City-scale	SAM-Road Baseline [18]	77.17	67.61

E. Evaluation Metrics

In this study, two primary metrics were used to evaluate the quality of road network extraction from satellite imagery: the TOPO F1-score [37] and the Average Path Length Similarity (APLS). The TOPO F1-score assesses the consistency of the predicted road network's topological structure compared to the ground truth. This metric measures the model's ability to reconstruct the topological relationships between nodes and edges, including in areas with complex intersections and road branches. The TOPO F1-score is calculated based on the precision and recall of correctly matched topological connections, in contrast to spurious and missing connections [38]. The mathematical definitions of spurious, missing, and the final TOPO F1-score are provided in Equations (2)–(4).

$$spurious = \frac{spurious_marbles}{spurious_marbles + matched_marbles} \quad (2)$$

$$missing = \frac{empty_holes}{empty_holes + matched_holes} \quad (3)$$

$$F1 = 2 \cdot \frac{precision \cdot recall}{precision + recall} = 2 \cdot \frac{(1 - spurious)(1 - missing)}{(1 - spurious) + (1 - missing)} \quad (4)$$

where,

- spurious* : Proportion of false predictions relative to correct ones
- spurious_marbles* : Road representation in the predicted graph
- matched_marbles* : Road segments in the predicted graph successfully matched with ground truth
- missing* : Proportion of missing predictions relative to expected predictions
- empty_holes* : Road representation in the ground truth graph
- matched_holes* : Road segments in the ground truth graph successfully matched with predictions
- F1* : Harmonic mean of precision and recall
- precision* : The ratio of correctly predicted road segments to all predicted segments
- recall* : The ratio of correctly predicted road segments to all ground truth segments

Equations (2), (3), and (4) describe the components used exclusively to compute the TOPO F1-score, namely *spurious*, *missing*, and their harmonic combination in the final TOPO metric. These formulas capture the topological accuracy of the extracted road graph in terms of connectivity between nodes and edges.

Meanwhile, APLS is used to evaluate the geometric accuracy of the extracted road paths. APLS measures the difference in the shortest path length between pairs of points on the ground truth graph and the predicted graph. The smaller the difference in path length, the higher the APLS value, indicating that the extracted paths spatially follow the true road structure [27]. These two metrics were chosen to provide a comprehensive evaluation: TOPO F1 focuses on the connectivity structure, while APLS captures the geometric accuracy of the extracted road paths. The APLS score is formally computed as shown in Equation (5).

$$C = 1 - \frac{1}{N} \sum \min \left\{ 1, \frac{|L(a,b) - L(a',b')|}{L(a,b)} \right\} \quad (5)$$

where,

- C : similarity score
- N : number of unique paths
- $L(a,b)$: length of path (a,b) in Graph Ground Truth
- $L(a',b')$: length of path (a',b') in Graph Predict
- $\min \{1, \dots\}$: maximum penalty

III. RESULTS

The experiments were organized into three groups: (1) gamma correction applied individually, (2) modified A* algorithm applied individually, and (3) a combination of both gamma correction and the modified A* algorithm. The experimental setup in this study was designed to evaluate the impact of gamma correction and the modified A* algorithm on the performance of SAM-Road in road network extraction.

In the initial phase, five gamma values (1.25, 1.5, 1.75, 2.0, and 2.2) were tested independently, without applying the A* algorithm, to observe their influence on segmentation quality and the resulting road graph structure. These values were selected based on prior studies demonstrating their effectiveness in enhancing image brightness and contrast [39]. The results of this experiment are summarized in Table 2.

Before applying any enhancement, the baseline performance of the SAM-Road Baseline [18] was recorded as 80.28 (TOPO F1-score) and 70.68 (APLS) for the SpaceNet dataset, and 77.17 (TOPO F1-score) and 67.61 (APLS) for the City-scale dataset. These baseline values serve as a reference for evaluating subsequent improvements. As shown in Table 2, gamma correction alone provides a noticeable enhancement in several configurations, particularly at $\gamma = 1.5$ for SpaceNet and $\gamma = 2.0$ for City-scale. Compared to the SAM-Road Baseline [18], these configurations improved TOPO F1-score by +0.33 and +1.39, and APLS by +0.23 and +1.12, respectively. This demonstrates that gamma correction contributes to better segmentation visibility and improved topological connectivity before applying the A* algorithm.

TABLE 2
PERFORMANCE COMPARISON BETWEEN THE BASELINE SAM-ROAD AND THE PROPOSED METHOD USING GAMMA CORRECTION

Dataset	Model	Gamma Value	TOPO F1-score	Δ TOPO F1-score	APLS	Δ APLS
SpaceNet	SAM-Road Baseline [18]	1.25	80.55	0.27	70.91	0.23
		1.5	80.61	0.33	69.74	-0.94
		1.75	80.32	0.04	69.98	-0.7
		2	80.31	0.03	69.20	-1.48
		2.2	79.87	-0.41	68.26	-2.42
City-scale	SAM-Road Baseline [18]	1.25	78.01	0.84	68.73	1.12
		1.5	78.26	1.09	67.19	-0.42
		1.75	78.34	1.17	67.68	0.07
		2	78.56	1.39	65.75	-1.86
		2.2	78.53	1.36	65.65	-1.96

Subsequently, a post-processing stage was conducted using the modified A* algorithm with three combinations of Min Graph Distance (16, 32, and 64 meters) and Max Straight Distance (8, 16, and 32 meters), adapted from the extended-line approach to handle occlusions. The evaluation of this algorithm alone is presented in Table 3.

TABLE 3
 EVALUATION OF THE MODIFIED A* APPLIED TO THE SAM-ROAD MODEL TO HANDLE DISCONNECTED ROADS WITHOUT VISUAL ENHANCEMENT

Dataset	Model	Min Graph Distance (m)	Max Straight Distance (m)	TOPO F1-score	Δ TOPO F1-score	APLS	Δ APLS
SpaceNet	SAM-Road Baseline [18]	16	8	80.22	-0.06	70.6	-0.08
		32	16	79.35	-0.93	70.71	0.03
		64	32	76.76	-3.52	69.12	-1.56
City-scale	SAM-Road Baseline [18]	16	8	77.29	0.12	68.69	1.08
		32	16	77.19	0.02	70.45	2.84
		64	32	76.31	-0.86	70.94	3.33

Finally, the third group of experiments applied gamma correction followed by the modified A* algorithm to assess the combined effect of both enhancements. The results of this integrated approach, as presented in Table 4, highlight both the best- and worst-performing configurations for each dataset based on TOPO F1-score and APLS metrics. This comparison allows a clearer understanding of how parameter choices impact the effectiveness of the combined method.

TABLE 4
 RESULTS OF THE PROPOSED HYBRID METHOD THAT INTEGRATES BOTH GAMMA CORRECTION AND THE MODIFIED A* ALGORITHM

Dataset	Model	Min G. / Max S. Dist.	Gamma Value	TOPO F1-score	Δ TOPO F1-score	APLS	Δ APLS
SpaceNet	Proposed (Combined: Gamma Correction + Modified A*)	Best					
		16m / 8m	1.5	80.59	0.31	69.79	-0.89
		32m / 16m	1.25	79.49	-0.79	71.17	0.49
		Worst					
		64m / 32m	2.2	75.41	-4.87	67.59	-3.09
		64m / 32m	2.2	75.41	-4.87	67.59	-3.09
City-scale	Proposed (Combined: Gamma Correction + Modified A*)	Best					
		16m / 8m	2	78.56	1.39	66.97	-0.64
		64m / 32m	1.75	76.84	-0.33	71.19	3.58
		Worst					
		64m / 32m	2.2	76.52	-0.65	70.19	+2.58
		16m / 8m	2.2	78.5	+1.33	66.39	-1.22

IV. DISCUSSION

Our experimental results demonstrate that optimal gamma values vary across metrics and datasets when addressing occlusion challenges. For the SpaceNet dataset, gamma 1.5 achieved the highest TOPO F1-score improvement (+0.33), while gamma 1.25 was optimal for APLS (+0.23). The City-scale dataset exhibited a similar pattern, with gamma 2.0 providing the best TOPO F1-score enhancement (+1.39) and gamma 1.25 remaining optimal for APLS (+1.12).

These findings reveal a fundamental trade-off between topological connectivity enhancement and geometric accuracy preservation. While gamma correction effectively improves road network topology structure, excessive gamma values lead to over-enhancement that compromises geometric precision. This observation aligns with previous studies indicating that aggressive preprocessing can degrade fine-grained geometric details crucial for accurate road detection.

Based on this analysis, we recommend employing gamma 1.25 for APLS optimization and gamma 1.5-2.0 for TOPO F1-score optimization, with the specific value selection guided by application priorities and dataset characteristics. As gamma increases, roads may appear more connected visually, supporting topological reconstruction, but this can lead to over-enhancement [40] and geometric inaccuracy. Figure 6 illustrates the visual impact of gamma correction on (a) SpaceNet and (b) City-scale, highlighting differences in enhancement levels across gamma values.

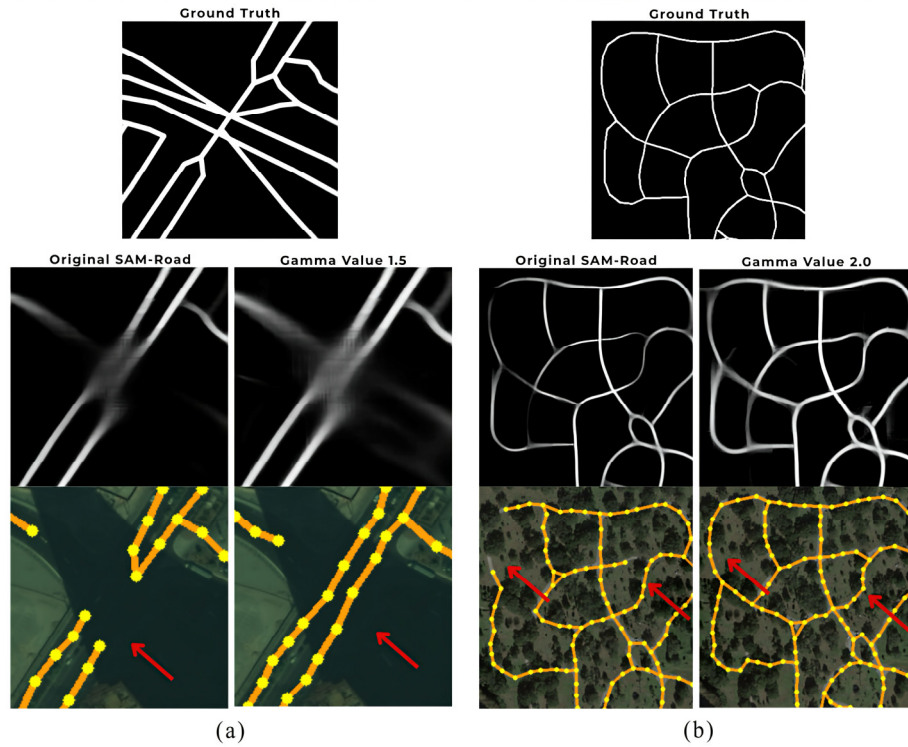


Fig. 6 Visualization of Gamma Correction on (a) SpaceNet and (b) City-scale.

For the modified A* algorithm, the results suggest that shorter distances are more suitable for SpaceNet, while longer distances enhance geometric accuracy in City-scale. Notably, the highest APLS (70.94) in City-scale was obtained at 64m/32m, although TOPO F1-score declined. These results confirm that the effectiveness of the modified A* algorithm depends on dataset characteristics. Larger parameters tend to benefit datasets with weak road connectivity while potentially harming datasets with well-established road structures. This again indicates a trade-off. Figure 4 displays how different configurations influence the extracted graph structure. Figure 7 visualizes the effect of the modified A* algorithm on (a) SpaceNet and (b) City-scale.

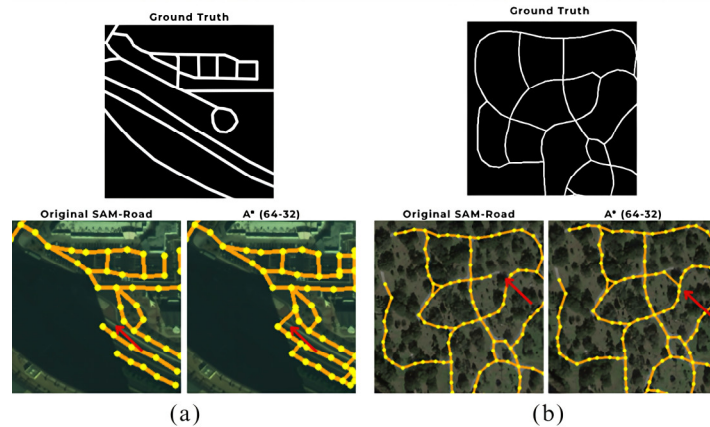


Fig. 7 Visualization of Modified A* on (a) SpaceNet and (b) City-scale.

The combination of gamma correction and the modified A* algorithm demonstrates varying effectiveness depending on dataset characteristics and evaluation priorities. On the SpaceNet dataset, gamma 1.5 with A* (16m, 8m) yields the highest TOPO score (80.59), while gamma 1.25 with A* (32m, 16m) performs best in APLS (71.17). For the City-scale dataset, gamma 2.0 with A* (16m, 8m) achieves the highest TOPO (78.56), and gamma 1.75 with

A* (64m, 32m) delivers the best APLS (71.19), with a +3.58 gain from baseline. This synergy confirms that gamma correction enhances visual clarity, and the modified A* reinforces topological continuity, making the combined method more effective, especially in highly fragmented scenarios. These patterns are visualized in Figure 5, showing how each parameter influences the final road network graph quality. Figure 8 illustrates the combined effect of gamma correction and the modified A* algorithm on (a) SpaceNet and (b) City-scale.

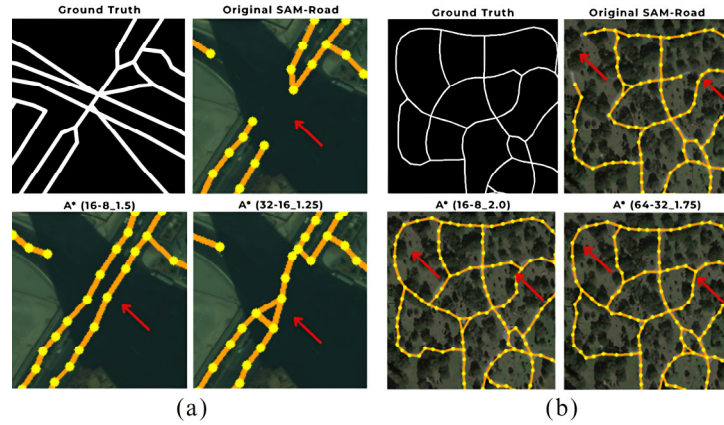


Fig. 8 Visualization of Gamma Correction and Modified A* on (a) Spacenet and (b) City-scale

TABLE 5
SUMMARY COMPARISON HIGHLIGHTING THE FINAL IMPROVEMENT OF THE PROPOSED HYBRID METHOD AGAINST THE SAM-ROAD
BASELIND [18]

Dataset	Model	Gamma Correction	Modified A*	TOPO F1-score	APLS
SpaceNet	SAM-Road Baseline [18]	V	V	80.28	70.68
	Proposed			80.59 (+0.31)	71.17 (+0.49)
City-scale	SAM-Road Baseline [18]	V	V	77.17	67.61
	Proposed			78.56 (+1.39)	71.19 (+3.58)

Table 5 presents the comparison between the SAM-Road Baseline [18] model and the best-performing configurations after applying Gamma Correction and the Modified A* algorithm. The results show that optimal parameter settings differ slightly between TOPO and APLS metrics across datasets. For the SpaceNet dataset, the best configuration (Gamma = 1.5, A* = 16–8) improved the TOPO F1-score from 80.28 to 80.59 (+0.31) and APLS from 70.68 to 71.17 (+0.49). Meanwhile, for the City-scale dataset, the best configuration (Gamma = 2.0, A* = 16–8 for TOPO and Gamma = 1.75, A* = 64–32 for APLS) increased the TOPO F1-score from 77.17 to 78.56 (+1.39) and APLS from 67.61 to 71.19 (+3.58). These consistent improvements show that Gamma Correction helps make road features more visible under difficult lighting conditions, while the Modified A* algorithm reconnects road segments that were separated due to occlusions. Compared to previous studies that mainly focused on improving segmentation accuracy or performing simple topological fixes, the proposed method provides a more complete approach by combining visual enhancement and topological reconstruction, resulting in better road extraction performance across different urban environments.

Relative to contemporary iterative graph-growing architectures such as RNGDet++ [41], the proposed method exhibits a distinct advantage in maintaining global network continuity, particularly within large-scale urban environments. While RNGDet++ achieves a higher TOPO F1-score of 82.51 on the SpaceNet dataset, this study's approach demonstrates superior performance in the APLS metric across both datasets, particularly on City-scale (71.19% vs. 67.76%). This suggests that the heuristic re-connection of disjointed segments using a modified A* algorithm is more effective at resolving long-range occlusions than agent-based traversal, which is prone to premature termination upon encountering dense obstacles or complex shadows.

Furthermore, when benchmarked against the emerging differentiable Bézier optimization model DOGE [42], the proposed method remains highly competitive. Although DOGE reports superior topological precision on SpaceNet (84.58) due to its continuous curve parameterization, this hybrid approach achieves a higher APLS on the City-scale

dataset (71.19% vs. 70.24%). Table 6 shows that both methods have complementary strengths across the two datasets. This inversion indicates a significant finding: while end-to-end differentiable optimization excels at capturing local shape fidelity and smoothness, the integration of robust post-processing algorithms like the modified A* provides a more reliable mechanism for ensuring the global routability and structural integrity required for practical geospatial and transportation applications. The proposed method thus offers a balanced trade-off between visual enhancement and topological reconstruction without the need for the complex vector ground-truth required by curve-based optimization models.

TABLE 6
PERFORMANCE COMPARISON OF THE PROPOSED HYBRID METHOD AGAINST RINGDET++ [41] AND DOGE [42] MODELS ON SPACENET AND CITY-SCALE DATSETS

Dataset	Model	TOPO F1-score	APLS
SpaceNet	RNGDet++ [41]	82.51	67.73
	DOGE [42]	84.58	73.48
	Proposed	80.59	71.17
City-scale	RNGDet++ [41]	78.44	67.76
	DOGE [42]	80.59	70.24
	Proposed	78.56	71.19

V. CONCLUSIONS

The research results show that each method tested for extracting road networks from satellite images has its own strengths and weaknesses. The baseline SAM-Road method performed well in terms of TOPO F1-score and APLS, producing relatively accurate road network extractions. However, it struggled with maintaining continuity, especially in areas affected by occlusion. When gamma correction was applied, it improved the visibility of road features in the segmentation masks produced by SAM-Road, leading to a better TOPO F1-score. This indicates that gamma correction helped the model identify parts of the road that were previously hidden or unclear due to occlusion, improving road continuity in those areas. As a result, the road network became more connected and accurate in terms of topology. However, this improvement in road continuity came with a decrease in APLS, showing that there is a trade-off between improving road continuity and maintaining geometric accuracy. The drop in APLS may be due to the geometric changes caused by gamma correction, which led to a mismatch between the extracted road network and the ground truth.

The modified A* algorithm, which was used to improve road network continuity, showed an increase in APLS. This suggests that A* is effective at filling in gaps where roads are disconnected due to occlusion. The algorithm worked well in dealing with occlusion, where parts of the road were hidden by objects and not detected in the initial extraction. By filling in these gaps, A* produced a more complete road network. However, when used without gamma correction, A* led to a decrease in TOPO F1-score because the additional paths created by A* sometimes resulted in connections that did not match the actual road structure. While A* was useful in handling occlusion, it was less effective in improving the overall accuracy of road network extraction.

The best results came from combining gamma correction with the modified A* algorithm. This combination effectively addressed the weaknesses of each method. Gamma correction improved the visibility of road features, while A* helped reconnect broken road segments. Together, they produced a more complete and connected road network. In areas affected by occlusion, such as shadows, trees, or buildings, this combination successfully overcame the challenges. Gamma correction clarified the segmentation mask, while A* filled in the hidden road segments, ensuring the road network stayed connected. The final result was a road network that was not only more complete in structure but also closer to the ground truth in terms of geometry than when using each method separately. Unlike the extended-line method, which is primarily effective for reconnecting linear road segments and often fails in areas with curves or intersections, this combined approach offers greater flexibility and robustness in reconstructing complex road geometries.

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REFERENCES

- [1] M. Thondoo, O. Marquet, S. Márquez, and M. J. Nieuwenhuijsen, "Small cities, big needs: Urban transport planning in cities of developing countries," *J Transp Health*, vol. 19, Dec. 2020, doi: 10.1016/j.jth.2020.100944.
- [2] P. Liang, W. Shi, Y. Ding, Z. Liu, and H. Shang, "Road extraction from high resolution remote sensing images based on vector field learning," *Sensors*, vol. 21, no. 9, May 2021, doi: 10.3390/s21093152.
- [3] T. Lobsang, F. Zhen, and S. Zhang, "Can urban street network characteristics indicate economic development level? Evidence from Chinese cities," *ISPRS Int J Geoinf*, vol. 9, no. 1, Dec. 2019, doi: 10.3390/ijgi9010003.
- [4] N. M. T. M. Mohamed and T. A. L. Aboul-Atta, "Roads networks investments as driver of regional development," *Journal of Umm Al-Qura University for Engineering and Architecture*, 2025, doi: 10.1007/s43995-025-00150-z.
- [5] S. Yang, O. Stepanchuk, O. Pylypenko, J. Ji, and A. Bieliatynskiy, "Scientific basis for improving the efficiency of urban street and road network operation," *Journal of Navigation*, vol. 76, no. 6, pp. 685–708, Nov. 2023, doi: 10.1017/S037346332400002X.
- [6] F. Ahmadzai, K. M. L. Rao, and S. Ulfat, "Assessment and modelling of urban road networks using Integrated Graph of Natural Road Network (a GIS-based approach)," *Journal of Urban Management*, vol. 8, no. 1, pp. 109–125, 2019, doi: 10.1016/j.jum.2018.11.001.
- [7] J. Kunkler and F. Kellner, "Sustainable City Evaluation Using the Database for Estimation of Road Network Performance," *Sustainability (Switzerland)*, vol. 15, no. 1, Jan. 2023, doi: 10.3390/su15010733.
- [8] H. Li, W. Huang, Y. Xu, W. Zhao, X. Zheng, and J. Yang, "Research on Strengthen the Development and Utilization of Geospatial Data to Release the Value of Data Elements," in *International Archives of the Photogrammetry, Remote Sensing and Spatial Information Sciences - ISPRS Archives*, International Society for Photogrammetry and Remote Sensing, Jul. 2025, pp. 913–919. doi: 10.5194/isprs-archives-XLVIII-G-2025-913-2025.
- [9] M. Buuveibaatar, K. Lee, and W. Lee, "Development of a Conceptual Data Model for 3D Geospatial Road Management Based on LandInfra Standard: A Case Study of Korea," *ISPRS Int J Geoinf*, vol. 11, no. 5, May 2022, doi: 10.3390/ijgi11050316.
- [10] Y. Deng, J. C. P. Cheng, and C. Anumba, "A framework for 3D traffic noise mapping using data from BIM and GIS integration," *Structure and Infrastructure Engineering*, vol. 12, no. 10, pp. 1267–1280, Oct. 2016, doi: 10.1080/15732479.2015.1110603.
- [11] I. Fabris-Rotelli *et al.*, "An informal road detection neural network for societal impact in developing countries," in *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences*, Copernicus GmbH, May 2022, pp. 267–274. doi: 10.5194/isprs-Annals-V-4-2022-267-2022.
- [12] R. Mahabir, A. Stefanidis, A. Croitoru, A. T. Crooks, and P. Agouris, "Authoritative and volunteered geographical information in a developing country: A comparative case study of road datasets in Nairobi, Kenya," *ISPRS Int J Geoinf*, vol. 6, no. 1, Jan. 2017, doi: 10.3390/ijgi6010024.
- [13] T. Y. D. Putra, Y. Sekimoto, and R. Shibasaki, "Toward the evolution of national spatial data infrastructure development in Indonesia," *ISPRS Int J Geoinf*, vol. 8, no. 6, Jun. 2019, doi: 10.3390/ijgi8060263.

- [14] V. P. Desai and H. Vala, "Survey on Methods of Road Extraction using Satellite Image," *Remote Sens (Basel)*, vol. 6, no. 9, pp. 9014–9033, Nov. 2014, doi: 10.3390/rs6099014.
- [15] S. Somvanshi, D. Kumar, and M. Kumari, "Trends of recent data, AI/ML approaches for geospatial AI in Earth observation towards sustainable development goals," 2025. [Online]. Available: <https://www.itu.int/en/journal/j-fet/Pages/default.aspx>
- [16] N. Netirith and M. Ji, "Analysis of the Efficiency of Transport Infrastructure Connectivity and Trade," *Sustainability (Switzerland)*, vol. 14, no. 15, Aug. 2022, doi: 10.3390/su14159613.
- [17] Y. Wei, Z. Wang, and M. Xu, "Road Structure Refined CNN for Road Extraction in Aerial Image," *IEEE Geoscience and Remote Sensing Letters*, vol. 14, no. 5, pp. 709–713, 2017, doi: 10.1109/LGRS.2017.2672734.
- [18] C. Hetang, H. Xue, C. Le, T. Yue, W. Wang, and Y. He, "Segment Anything Model for Road Network Graph Extraction," Apr. 2024, [Online]. Available: <http://arxiv.org/abs/2403.16051>
- [19] Z. Liu, W. Jin, and Y. Mu, "Graph-based boosting algorithm to learn labeled and unlabeled data," *Pattern Recognit*, vol. 106, p. 107417, 2020, doi: 10.1016/j.patcog.2020.107417.
- [20] Y. Zhao *et al.*, "A deeply supervised vertex network for road network graph extraction in high-resolution images," *International Journal of Applied Earth Observation and Geoinformation*, vol. 133, Sep. 2024, doi: 10.1016/j.jag.2024.104082.
- [21] P. Yin *et al.*, "Towards Satellite Image Road Graph Extraction: A Global-Scale Dataset and A Novel Method," Nov. 2024.
- [22] D. Feng, X. Shen, Y. Xie, J. Hu, and Y. Liu, "Efficient occluded road extraction from high-resolution remote sensing imagery," *Remote Sens (Basel)*, vol. 13, no. 24, Dec. 2021, doi: 10.3390/rs13244974.
- [23] F. M. A. Farzana and E. F. I. Rani, "Technically Improved Image and Video Enhancement using Adaptive Gamma Correction with Weighing Distribution Based Contrast Enhancement Techniques," vol. 3, no. 1, pp. 50–57, 2019.
- [24] M. Zhang, B. Wang, S. Yang, Q. Liu, and Y. Wang, "kLCRNet: Fast Road Network Extraction via Keypoint-Driven Local Connectivity Exploration," *IEEE J Sel Top Appl Earth Obs Remote Sens*, vol. 18, pp. 12074–12089, 2025, doi: 10.1109/JSTARS.2025.3564060.
- [25] D. Zhao *et al.*, "A Study of the Improved A* Algorithm Incorporating Road Factors for Path Planning in Off-Road Emergency Rescue Scenarios," *Sensors*, vol. 24, no. 17, Sep. 2024, doi: 10.3390/s24175643.
- [26] Y. He, R. Garg, and A. R. Chowdhury, "TD-Road: Top-Down Road Network Extraction with Holistic Graph Construction."
- [27] A. Van Etten, D. Lindenbaum, and T. M. Bacastow, "SpaceNet: A Remote Sensing Dataset and Challenge Series," Jul. 2018.
- [28] S. He, F. Bastani, S. Jagwani, and M. Alizadeh, "Sat2Graph : Road Graph Extraction," pp. 20–22, 2020.
- [29] N. F. Sahib and Z. A. Hashim, "Contrast Image Enhancement By Gamma Correction," *Computer Engineering and Intelligent Systems*, vol. 9, no. 7, pp. 55–66, 2018.
- [30] M. Mahamdioua and M. Benmohammed, "New Mean-Variance Gamma Method for Automatic Gamma Correction," *International Journal of Image, Graphics and Signal Processing*, vol. 9, no. 3, pp. 41–54, 2017, doi: 10.5815/ijigsp.2017.03.05.
- [31] K. Somasundaram and P. Kalavathi, "Medical Image Contrast Enhancement Based ON Gamma Correction," *International Journal of Knowledge Management and e-Learning*, vol. 3, no. 1, pp. 15–18, 2011.
- [32] Y. Yan and B.- Dublin, "Research on the A Star Algorithm for Finding Shortest Path," 2023.
- [33] G. Mattyus, W. Luo, and R. Urtasun, "DeepRoadMapper: Extracting Road Topology from Aerial Images," *Proceedings of the IEEE International Conference on Computer Vision*, vol. 2017-Octob, pp. 3458–3466, 2017, doi: 10.1109/ICCV.2017.372.
- [34] D. Rachmawati and L. Gustin, "Analysis of Dijkstra's Algorithm and A* Algorithm in Shortest Path Problem," *J Phys Conf Ser*, vol. 1566, no. 1, 2020, doi: 10.1088/1742-6596/1566/1/012061.
- [35] C. Hetang, H. Xue, C. Le, T. Yue, W. Wang, and Y. He, "Segment Anything Model for Road Network Graph Extraction," Mar. 2024, doi: 10.1109/CVPRW63382.2024.00262.
- [36] X. Sun, "On Efficient Variants of Segment Anything Model: A Survey," 2024.
- [37] S. He *et al.*, "RoadRunner: Improving the precision of road network inference from GPS trajectories," in *GIS: Proceedings of the ACM International Symposium on Advances in Geographic Information Systems*, Association for Computing Machinery, Nov. 2018, pp. 3–12. doi: 10.1145/3274895.3274974.
- [38] J. Biagioni and J. Eriksson, "Inferring road maps from global positioning system traces," *Transp Res Rec*, no. 2291, pp. 61–71, Dec. 2012, doi: 10.3141/2291-08.
- [39] M. Naufal, H. Al Azies, and R. M. Brilianto, "Enhanced Brain Tumor Classification Through Gamma Correction in Deep Learning," vol. 13, pp. 2348–2358, 2024.
- [40] S. Rahman, M. M. Rahman, M. Abdullah-Al-Wadud, G. D. Al-Quaderi, and M. Shoyaib, "An adaptive gamma correction for image enhancement," *EURASIP J Image Video Process*, vol. 2016, no. 1, pp. 1–13, 2016, doi: 10.1186/s13640-016-0138-1.
- [41] Z. Xu, Y. Liu, Y. Sun, M. Liu, and L. Wang, "RNGDet++: Road Network Graph Detection by Transformer with Instance Segmentation and Multi-scale Features Enhancement," Mar. 2023, [Online]. Available: <http://arxiv.org/abs/2209.10150>
- [42] J. Sun, J. Lu, J. Yin, Y. Xu, Y. Li, and Y. Guo, "DOGE: Differentiable Bezier Graph Optimization for Road Network Extraction," Nov. 2025, [Online]. Available: <http://arxiv.org/abs/2511.19850>.

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